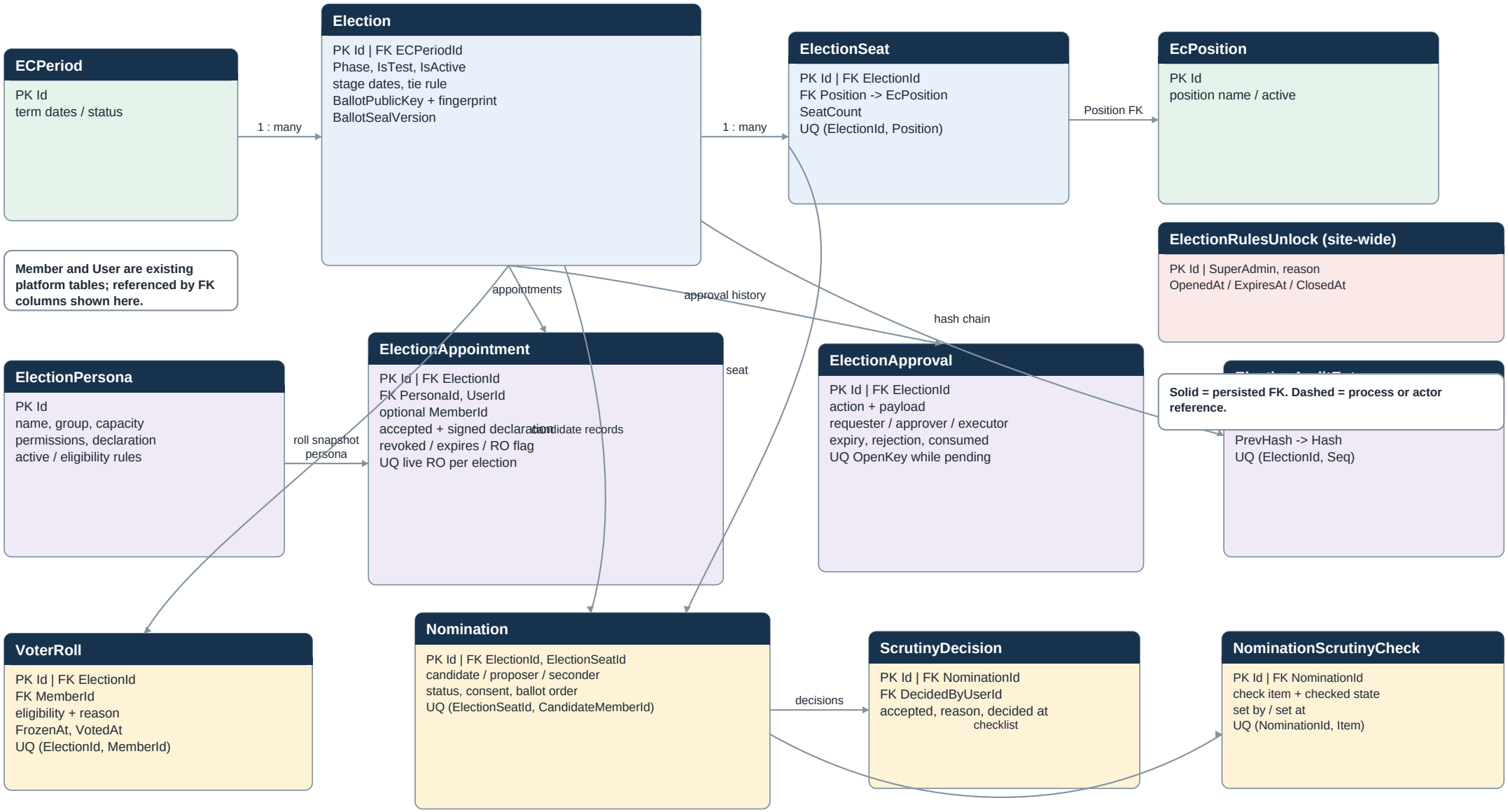
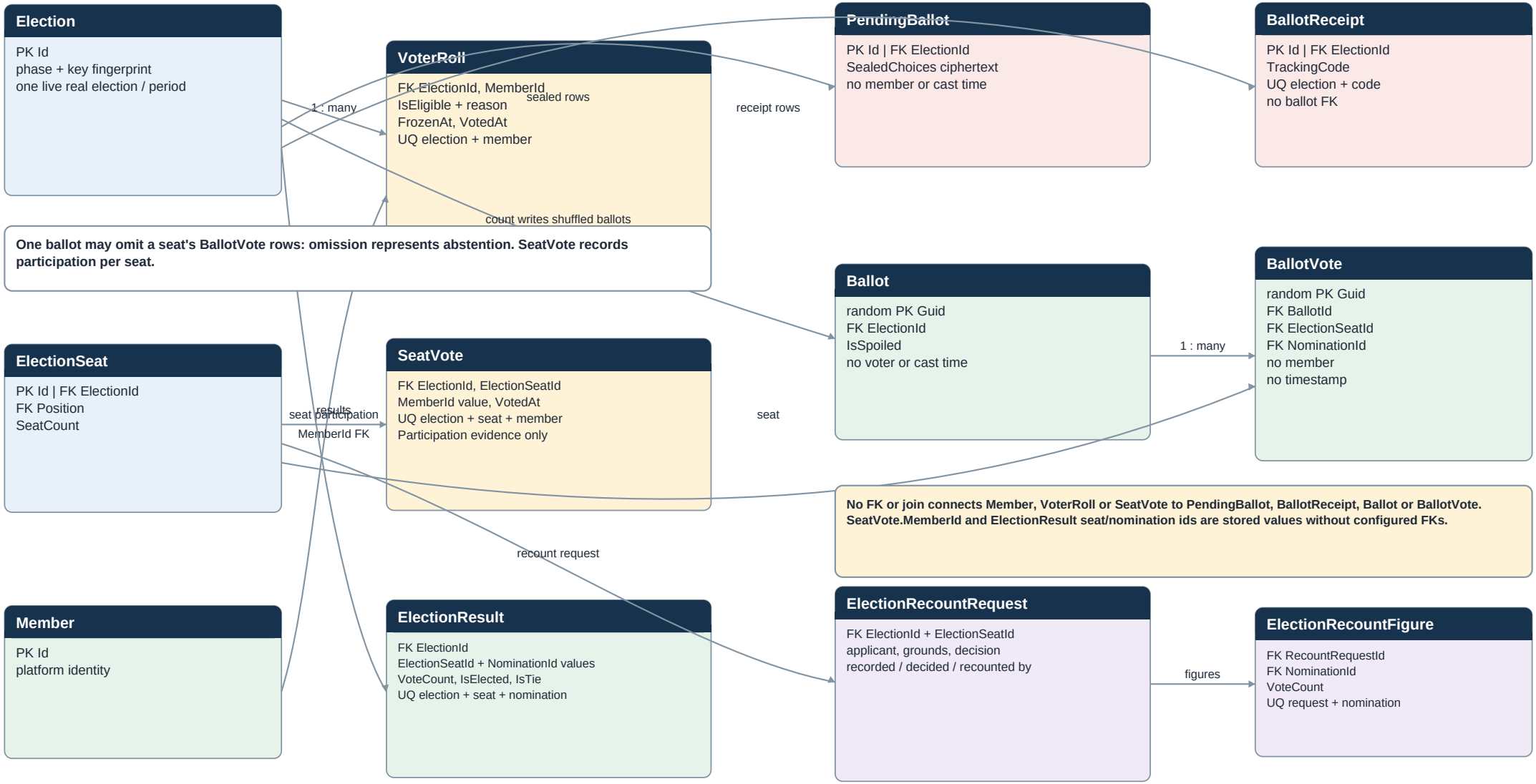
Election ERD | 1. Governance, eligibility and nominations

Persisted election setup and candidate preparation. PK/FK labels reflect current domain and EF configuration.



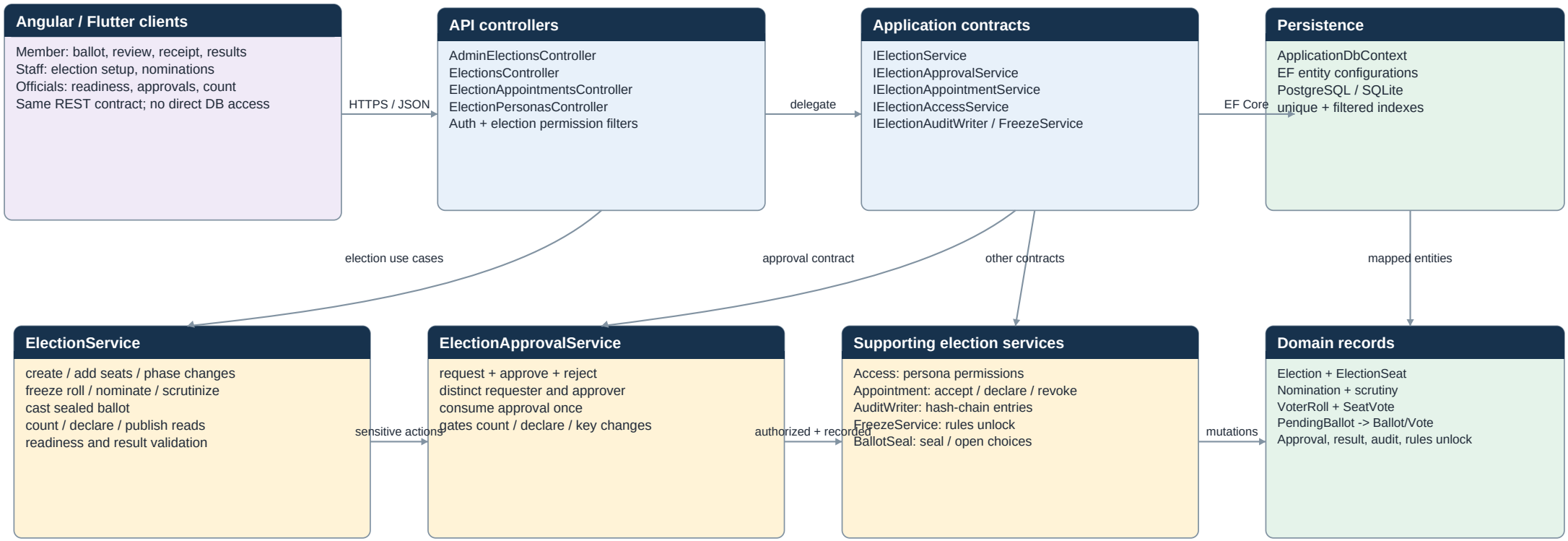
Election ERD | 2. Participation, sealed ballots and outcomes

Ballot secrecy depends on keeping voter-side evidence separate from anonymous counted choices.



Election class diagram | Components and responsibilities

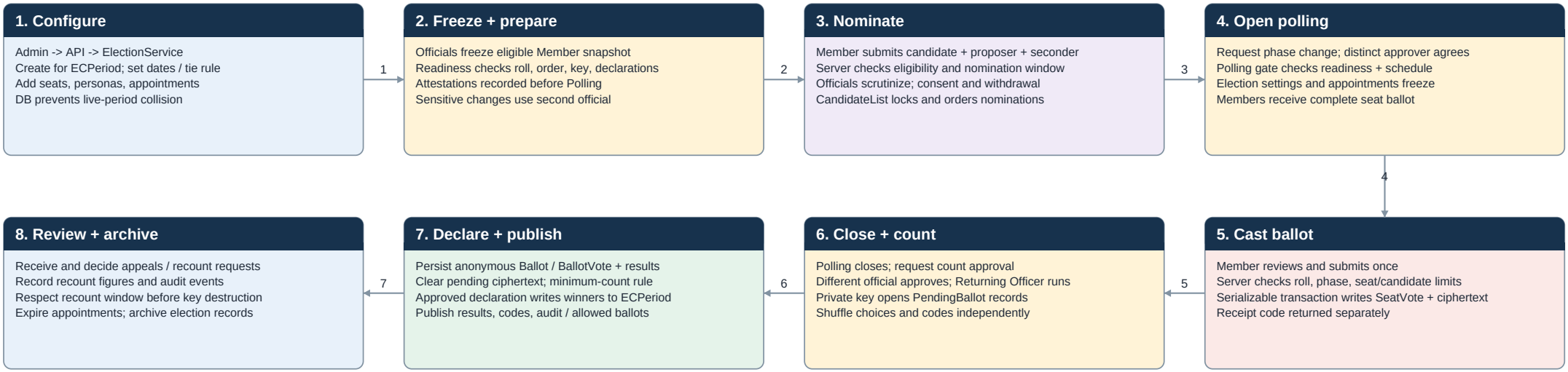
Client screens call REST endpoints; controllers delegate to application contracts; services enforce election rules and persist domain records.



Privacy invariant: identified participation is stored in VoterRoll / SeatVote. Sealed choices wait in PendingBallot. At count, the service opens and shuffles ballots and tracking codes independently, then writes anonymous Ballot / BallotVote and results. No class relationship links voter identity to counted choices. The private key is supplied for the count and is not stored on Election.

Election class diagram | Full lifecycle and control gates

End-to-end sequence from election creation through appeals and archive. Approval is a separate action from execution.



Control gates and data transitions • Phase changes are validated by ElectionService; sensitive actions are requested and approved by distinct officials through ElectionApprovalService. • VoterRoll is frozen before the election progresses. A vote is one full submission, while SeatVote proves participation separately. • PendingBallot holds ciphertext only. At count, the service decrypts using the Returning Officer's supplied private key, shuffles ballots and tracking codes separately, writes anonymous counted records and removes pending rows. • Count and declaration are separately approved. A failed readiness gate prevents Polling; a failed count leaves the election undeclared. Audit entries form a per-election hash chain.